SPESIFIKASI SISTEM · SYSTEM SPECIFICATION

e-HadirGo

Kehadiran Staf Luar Pejabat dengan Pengesahan Lokasi

Jabatan Perikanan Sabah

Versi	1.0	**Tarikh**	6 November 2026
Status	Draf untuk semakan	**Platform**	Web + PWA
Bahasa	Bahasa Melayu / English	**Peringkat**	Fasa reka bentuk

Dokumen ini menerjemahkan catatan keperluan bertulis tangan bertarikh 6 November 2026 kepada spesifikasi yang boleh dibina: peranan, aliran sistem, peraturan kehadiran, model data, kontrak API dan pelan pelaksanaan.

This document translates the handwritten requirements of 6 November 2026 into a build-ready specification: roles, system flows, attendance rules, data model, API contract and delivery plan.

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Ringkasan Eksekutif — Executive Summary

e-HadirGo is a closed, staff-only attendance system for Jabatan Perikanan Sabah. It answers one question with evidence: when a staff member is working outside the office, were they at the right place, at the right time, and if not, why not?

The design rests on five principles:

1. **Evidence, not punishment.** An out-of-area or late check-in is recorded with coordinates, distance and a written reason — flagged for the supervisor rather than silently blocked. Rural GPS in Sabah is often poor; the system must not punish a weak signal, but it must preserve the record.
2. **One tap in the field.** Check-in is a single action from a phone, reachable from a Telegram message or the PWA. Any extra step is a step staff will skip.
3. **Closed by construction.** Only accounts with a verified @sabah.gov.my address activate automatically. Every account starts as staff; supervisor rights are granted, never claimed.
4. **Free where it matters.** Telegram for reminders, Gmail for the formal record, browser geolocation for position. No per-message cost, no WhatsApp dependency, no Google Maps bill.
5. **Bilingual by default.** Every screen, email and report exists in Bahasa Melayu and English, switchable at any time.

> **Ringkasan keputusan:** Email + password login · staff by default · supervisor granted by super admin · pin + radius location · any-time early check-in · late after 15 minutes with a required reason · required catatan outside the radius · Telegram + Gmail only · BM/EN · web + PWA.

1. Pengenalan — Overview

1.1 Tujuan dokumen

This document is the reference for building e-HadirGo. It converts the handwritten note of 6 November 2026 — system for staff attendance outside the office by masa and tarikh, gov-email sign-up, department selection, email reminders and location proof — into unambiguous requirements, a data model and an API contract. Where the note was silent, a decision has been recorded in Section 14 rather than left implicit.

1.2 Latar belakang

Staff of Jabatan Perikanan Sabah regularly work away from their desks: jetties, hatcheries, enforcement patrols, district offices and field stations across five districts. Attendance for that work is currently recorded on paper or in chat messages, which proves neither the time nor the place. Supervisors cannot tell a genuine field visit from a claim, and monthly reporting requires manual reconstruction.

e-HadirGo closes that gap with a small, auditable loop: a supervisor defines an assignment (who, when, where), the staff member checks in from their phone at that place, and the system records time, coordinates and any exception with a written explanation.

1.3 Skop

Dalam skop — In scope	Di luar skop — Out of scope
Responsive web application, installable as a PWA	Native iOS/Android apps in an app store
Sign-up, login, password reset, session management	Single sign-on / LDAP / HR system integration
Role model: staff, supervisor (additive), super admin	Granular custom roles or per-unit delegated admins
Assignment creation with map pin and radius	Route tracking, mileage claims, fleet management
Check-in / check-out with location and time validation	Continuous or background GPS tracking
Telegram and Gmail notifications	WhatsApp, SMS, or voice calls
Bilingual UI, email and PDF reports	Additional languages beyond BM and EN
CSV/PDF reporting and audit trail	Payroll, overtime or disciplinary workflow

1.4 Glosari

Istilah	English	Maksud
Tugasan	Assignment	A defined field task: staff, date, time window, location, radius, purpose
Kehadiran	Attendance	The record created by a check-in and closed by a check-out
Daftar masuk	Check-in	Staff confirms presence at the assignment location
Daftar keluar	Check-out	Staff closes the attendance record
Luar kawasan	Outside radius	Device position is farther from the pin than the allowed radius
Lewat	Late	Check-in more than 15 minutes after the registered start time
Catatan	Note / reason	Free-text justification required for exceptions
Penyelia	Supervisor	Staff member granted the right to create assignments and review records
Lokasi tetap	Saved location	A reusable pin maintained by the admin (jetty, office, station)
Daerah / Jabatan / Bahagian	District / Department / Division	The three-level organisational unit

2. Pengguna & Peranan — Users and Roles

2.1 Definisi peranan

Peranan	English	Kebolehan	Cara diperoleh
Staf	Staff	Records their own attendance, sees their assignments, manages their profile and notification channels.	Automatic on sign-up
Penyelia	Supervisor	Everything a staff member can do, plus: create and edit assignments, view the live map, review exceptions, approve justifications, export unit reports.	Granted by super admin, additive to staff
Super Admin	Super Admin	Everything a supervisor can do, plus: manage users and roles, organisational units, saved locations, message templates, delivery logs and the audit trail.	Seeded account, maximum two holders

> **Model peranan:** A supervisor is a staff member who has been given an extra role, not a separate account. The account holds a list of roles — ["staf", "penyelia"] — and the interface simply reveals the additional menus. One person can therefore supervise a team and still check in to their own field work.

2.2 Matriks kebolehan

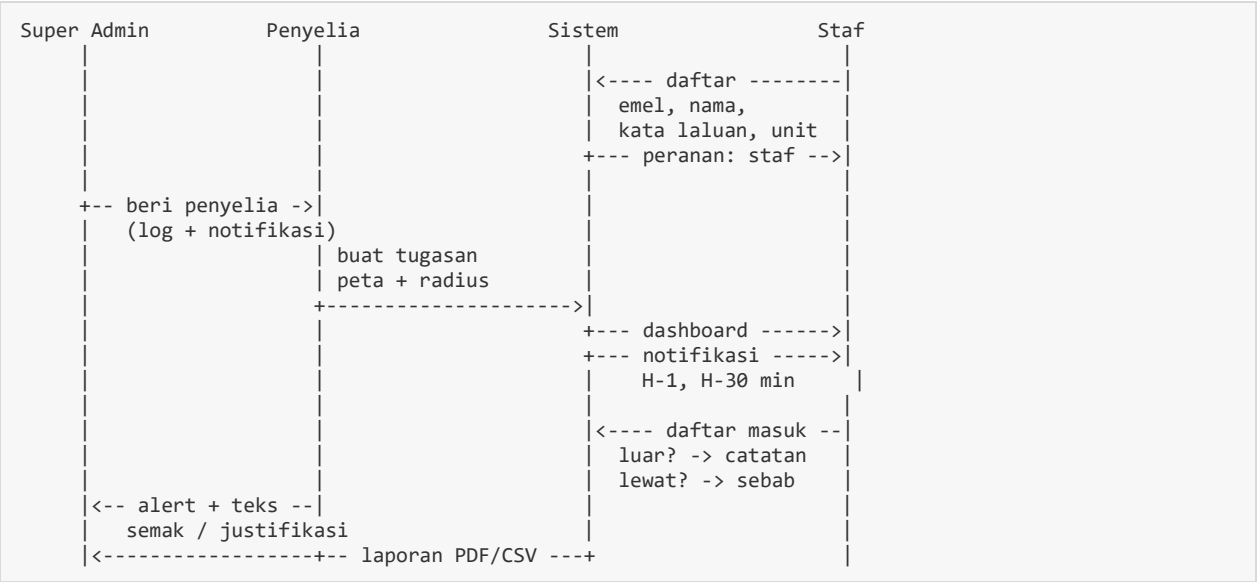
Kebolehan	Staf	Penyelia	Super Admin
Daftar akaun sendiri	Ya	Ya	Ya
Log masuk dengan emel + kata laluan	Ya	Ya	Ya
Lihat dashboard & tugasan sendiri	Ya	Ya	Ya
Daftar masuk / keluar	Ya	Ya	Ya
Cipta dan sunting tugasan	—	Ya	Ya
Tetapkan lokasi, radius dan tetingkap masa	—	Ya	Ya
Peta langsung semua staf	—	Ya	Ya
Semak pengecualian dan justifikasi	—	Ya	Ya
Laporan agregat unit	—	Ya	Ya
Urus pengguna dan beri peranan penyelia	—	—	Ya
Urus unit, lokasi tetap, templat mesej	—	—	Ya
Lihat log penghantaran dan audit	—	—	Ya

2.3 Peraturan pemberian peranan

- **Default.** Every new account is created with the role staf. There is no role selector on the sign-up form.
- **Grant.** Only a super admin can grant or revoke penyelia, from Pengguna & Peranan. The change takes effect at the next page load and is written to the audit log.
- **Notify.** The user is notified in-app and on their preferred channel when a role is added or removed.
- **Self-assignment.** A supervisor may create an assignment for themselves. This is permitted by decision D-04; the record stores both staff_id and supervisor_id, and the interface labels self-assigned records so a reviewer can see them at a glance.

3. Aliran Sistem — System Flows

3.1 Gambaran keseluruhan



3.2 Aliran A — Pendaftaran dan pengaktifan

#	Pelaku	Tindakan	Respons sistem
1	Staf	Membuka aplikasi dan memilih Daftar	Memaparkan borang: emel, nama, kata laluan, daerah, jabatan, bahagian
2	Staf	Mengisi dan menghantar borang	Mengesahkan format emel, domain pakai buang dan rekod MX
3	Sistem	Menghantar OTP 6 digit ke emel	Menyimpan akaun sebagai belum disahkan selama 10 minit
4	Staf	Memasukkan OTP	Menandakan emel disahkan, menetapkan peranan staf, akaun aktif
5	Sistem	Menyediakan onboarding	Menunjukkan panduan pautan Telegram dan pilihan bahasa
6	Staf	Memilih saluran dan menekan pautkan	Menyimpan channel_id selepas pengesahan bot; menghantar mesej ujian

If the email domain is not @sabah.gov.my, or the domain has no MX record, the form shows the verification warning described in 4.1. The account can still be completed with the dummy-data confirmation, and is flagged is_dummy for the admin.

3.3 Aliran B — Tugasan dan notifikasi

#	Pelaku	Tindakan	Respons sistem
1	Penyelia	Menekan Tugasan Baharu	Memaparkan peta, carian lokasi, pemilih staf dan medan masa
2	Penyelia	Meletakkan pin dan memilih radius	Mengira liputan, memaparkan bulatan, mengesahkan pin berada dalam Sabah
3	Penyelia	Menyimpan tugasan	Mencipta rekod berstatus Dijadualkan, mengesan pertindihan jadual staf
4	Sistem	Menghantar notifikasi segera	Telegram/emel: tugasan baharu dengan butang Daftar Masuk
5	Sistem	Menjadualkan peringatan	H-1 hari dan H-30 minit sebelum masa mula, mengikut saluran pilihan
6	Staf	Melihat kad pada dashboard	Kad Hari Ini / Akan Datang dengan lokasi, masa, radius dan status

3.4 Aliran C — Daftar masuk dan daftar keluar

#	Pelaku	Tindakan	Respons sistem
1	Staf	Menekan Daftar Masuk	Meminta kebenaran lokasi jika belum diberi; membaca GPS
2	Sistem	Mengira jarak ke pusat	Haversine ke pin; menyimpan latitud, longitud, ketepatan, jarak
3	Sistem	Menyemak masa	Awal / Tepat Masa / Lewat mengikut jadual 4.5
4	Sistem	Menilai syarat tambahan	Di luar radius memerlukan catatan; lewat memerlukan sebab
5	Staf	Menghantar borang	Menyimpan rekod, memaparkan resit dan status
6	Sistem	Memaklumkan penyelia	Alert segera untuk luar kawasan, lewat atau selepas tamat
7	Staf	Menekan Daftar Keluar	Menutup rekod, mengira tempoh, menyimpan lokasi akhir
8	Sistem	Menutup automatik	Selepas masa tamat + 30 minit tanpa daftar keluar

3.5 Aliran D — Semakan dan laporan

#	Pelaku	Tindakan	Respons sistem
1	Sistem	Mengumpul pengecualian	Luar kawasan, lewat, tidak hadir, daftar keluar tertinggal
2	Penyelia	Membuka Semakan	Senarai kes dengan peta, jarak, ketepatan dan teks staf
3	Penyelia	Menerima atau menolak justifikasi	Menyimpan keputusan, catatan penyelia dan cap masa
4	Penyelia	Mengeksport laporan	PDF/CSV mengikut tarikh, unit, staf dan status
5	Sistem	Mengarkib	Rekod kekal boleh audit; laporan bulanan dijana automatik

3.6 Status dan bendera

Status tugasan

Status	Bermakna	Peralihan seterusnya
Dijadualkan	Tugasan disimpan, belum bermula	Dalam tindakan / Dibatalkan
Dalam tindakan	Staf telah daftar masuk	Selesai / Tidak hadir
Selesai	Daftar keluar atau tutup automatik	—
Tidak hadir	Tetingkap tamat tanpa daftar masuk	Semakan penyelia
Dibatalkan	Dibatalkan oleh penyelia	—

Bendera rekod kehadiran

The attendance record keeps one operational status plus independent flags, so reporting can separate a late-but-present record from an on-time-but-elsewhere record without ambiguous enums.

Medan	Jenis	Maksud
status	enum	Hadir / Tidak hadir
is_early	boolean	Daftar masuk sebelum masa mula
is_late	boolean	Daftar masuk lebih 15 minit selepas masa mula
late_reason	text, wajib jika is_late	Sebab lewat, dilihat oleh penyelia
outside_radius	boolean	Jarak melebihi radius yang ditetapkan
outside_note	text, wajib jika outside_radius	Catatan lokasi
distance_m	integer	Jarak sebenar ke pusat tugasan
accuracy_m	integer	Ketepatan GPS yang dilaporkan peranti

4. Keperluan Fungsian — Functional Requirements

4.1 Pendaftaran — Sign-up

Sign-up requires exactly five fields. There is no IC number, no phone number and no role selector; a phone number can be added later from Tetapan.

Medan	Jenis	Peraturan
Emel	text	Wajib. Unik. Disahkan melalui MX + OTP. Domain @sabah.gov.my diutamakan
Nama penuh	text	Wajib. 3–120 aksara, seperti dalam rekod perjawatan
Kata laluan	password	Wajib. Minimum 8 aksara, satu huruf dan satu nombor

Medan	Jenis	Peraturan
Daerah	select	Wajib. Kota Kinabalu, Sandakan, Tawau, Lahad Datu, Kudat
Jabatan	select	Wajib. Lalai: Jabatan Perikanan Sabah
Bahagian	select	Wajib. Bergantung kepada daerah yang dipilih

Pengesahan emel

- **Langkah 1.** Syntax and length, trimmed and lower-cased before comparison.
- **Langkah 2.** Disposable and known test domains are rejected (mailinator, tempmail, example.com, dummy.test).
- **Langkah 3.** A DNS MX lookup confirms the domain can receive mail. No MX means the address is very likely dead.
- **Langkah 4.** A 6-digit OTP is sent and must be entered within 10 minutes. This is the only true proof that the mailbox works.

> **Laluan data dummy:** If MX fails or the domain is a known test domain, the screen shows a warning — "Emel tidak dapat disahkan. Domain tiada rekod MX." — with two actions: Tukar emel, or Teruskan dengan data dummy. Confirming sets email_verified = false and is_dummy = true, and the account is excluded from production reports until the address is corrected.

A valid @sabah.gov.my address activates the account immediately after OTP. There is no super-admin approval queue — the domain is the gate.

4.2 Log masuk — Login

Login is email plus password only. The system loads the user's roles from the database; the user never selects a role. Failed attempts are rate-limited to five per fifteen minutes per email and per IP. A successful login can be remembered on a trusted device for 30 days. Forgotten passwords are reset by a single-use link sent to the registered email.

4.3 Dashboard staf

The staff dashboard is the front door and must answer three questions in one screen: what am I doing today, where, and have I checked in?

- Kad Hari Ini — assignments whose date is today, with location name, radius, time window, status pill and a prominent Daftar Masuk button.
- Kad Akan Datang — the next seven days, grouped by date.
- Rekod Terkini — the last five attendance records with status flags.
- A banner appears if a notification channel is unlinked, prompting the user to complete the guide in Section 5.2.

An assignment appears on this dashboard the moment the supervisor saves it. There is no publish or approval step, and the notification fires in the same transaction.

4.4 Penetapan lokasi oleh penyelia

The supervisor defines one point and one circle. That pair is the entire geofence.

Kaedah	Cara	Bila digunakan
Carian	Taip nama tempat atau alamat; pin jatuh dan nama diisi automatik	Kes biasa

Kaedah	Cara	Bila digunakan
Ketik / seret peta	Ketik mana-mana titik atau seret pin; bulatan mengikut	Melaraskan hasil carian
Guna lokasi saya	GPS penyelia semasa, untuk tugasan di lokasi sebenar	Penyelia berada di tapak
Lokasi tetap	Senarai yang diselenggara admin: jeti, pejabat daerah, stesen, hatcheri	Lokasi berulang
Koordinat manual	Taip latitud dan longitud	Kawasan pedalaman tanpa alamat

Peta dan bulatan

The supervisor sees a live Leaflet map, not a static image. The pin is draggable and the circle is always centred on it. Selecting a radius resizes the circle immediately and updates the coverage readout (for example 250 m · 0.196 km²). A Lihat sebagai staf toggle previews exactly what the staff member will see.

Kawalan	Nilai	Nota
Radius pratetap	100 m / 250 m / 500 m / 1 km	Lalai 250 m, boleh diubah per tugasan
Radius tersuai	50 m – 5 km	Minimum 50 m untuk mengelak bunyi GPS
Nama lokasi	Wajib	Staf mesti nampak nama manusia, bukan koordinat
Semakan sempadan	Amaran lembut	Pin di luar Sabah disyaki salah taip

> **Reka bentuk peta:** Interactive map on the supervisor dashboard and the assignment form; a static map snapshot (one tile request) on the staff check-in screen and inside the PDF report. This keeps map tile usage inside the free tier while preserving the visual proof of the circle.

4.5 Peraturan masa — Time rules

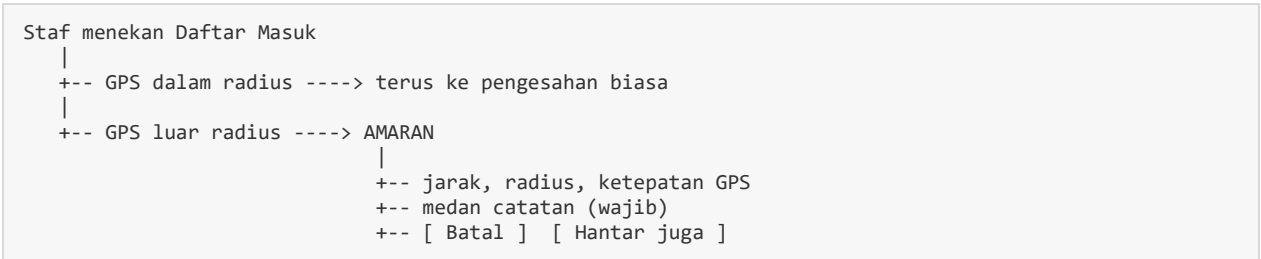
Check-in is allowed at any time before the registered start. There is no minimum lead time.

Masa daftar masuk	Status	Syarat tambahan
Sebelum masa mula (bila-bila masa)	Awal	Tiada. Tidak dikira lewat
Masa mula hingga +15 minit	Tepat Masa	Tiada
+15 minit hingga masa tamat	Lewat	Sebab wajib — untuk integriti kendiri
Selepas masa tamat	Lewat (luar tempoh)	Sebab wajib, dihantar kepada penyelia
Tiada daftar masuk	Tidak Hadir	Eskalasi automatik kepada penyelia

Every late record carries a written reason, not only the ones past the end time. The reason is shown to the supervisor beside the map and is reproduced in the monthly report.

4.6 Alert lokasi sebelum hantar

The location check is a two-step confirmation. It warns, it does not block — because a blocked legitimate check-in is worse than a flagged one.



If location permission is denied, or reported accuracy is worse than 100 m, a different message appears — "Lokasi tidak tepat, cuba di luar bangunan" — with a retry action and the same fallback path.

4.7 Daftar keluar

Check-out reuses the same location check and closes the record, computing duration. A record left open is closed automatically 30 minutes after the end time and flagged auto_closed, so a forgotten check-out never inflates hours.

4.8 Semakan penyelia

Semakan lists every exception for the supervisor's units: outside radius, late, late past end, no-show and auto-closed. Each row opens a detail view with the map, the distance, the reported accuracy, the staff member's text and two actions — Terima justifikasi or Tolak justifikasi — each requiring a short supervisor note. Decisions are timestamped and audited.

4.9 Tetapan

Bahagian	Boleh diubah	Tingkah laku
Profil	Nama, telefon, jawatan	Segera
Profil	Emel	Perlu pengesahan OTP semula
Profil	Daerah / Jabatan / Bahagian	Segera, dengan notifikasi kepada super admin
Notifikasi	Saluran utama dan sandaran	Segera selepas pautan disahkan
Bahasa	Bahasa Melayu / English	Segera, disimpan pada profil
Keselamatan	Kata laluan, peranti dipercayai	Segera
Privasi	Persetujuan lokasi, notis penyimpanan	Segera

> **Perubahan unit:** Instant, as decided, but never silent. The super admin receives an in-app and Telegram notification — "Unit berubah: Sandakan / Akuakultur → Kota Kinabalu / Pengurusan" — and an audit entry records the previous and new values.

4.10 Pentadbiran

- Pengguna & Peranan — search, view, grant or revoke penyelia, deactivate accounts.
- Unit — maintain daerah, jabatan and bahagian, and their combinations.
- Lokasi Tetap — the saved-location list used by supervisors.
- Templat Mesej — bilingual Telegram and email templates with variable preview.
- Log Penghantaran — every notification with channel, status and error text.
- Audit — who changed what, when, with before and after values.

4.11 Laporan

Reports filter by date range, unit, staff and status, and export to CSV or PDF. The PDF embeds a map snapshot with the pin and circle for each assignment, the staff member's reason text where applicable, and the supervisor's decision. A monthly recap per unit is generated automatically on the first working day of the following month.

5. Notifikasi

5.1 Saluran

Saluran	Kegunaan	Kos	Nota
Telegram	Peringatan, notifikasi segera, butang Daftar Masuk	Percuma sepenuhnya	Bot API: ~30 mesej/s keseluruhan, ~1 mesej/s per chat. Tiada kuota bulanan
Gmail	OTP, rekod rasmi, laporan, salinan notifikasi	Percuma dengan had	~500 penerima/hari untuk Gmail peribadi, ~2,000/hari untuk Google Workspace

> **Mengapa tiada WhatsApp:** There is no reliable free WhatsApp API. Meta Cloud API requires business verification and pre-approved templates, with authentication templates billed; Twilio charges per message; unofficial gateways breach WhatsApp's terms and risk a ban. Telegram plus Gmail covers the same need at zero cost. The notification layer is written behind a provider interface so WhatsApp can be added later without touching attendance logic.

5.2 Panduan pautan saluran

A staff member who selects Telegram cannot leave the screen until the link is verified. Each channel shows a status badge: Dipautkan, Belum dipautkan or Gagal.

Saluran	Langkah	Pengesahan
Telegram	1. Tekan Pautkan Telegram 2. Buka pautan t.me/<bot>?start=<token> 3. Tekan Start 4. Bot membalas akaun dipautkan	Mesej bot diterima dan chat_id disimpan
Gmail	1. Tekan Hantar OTP 2. Semak peti masuk 3. Masukkan 6 digit	OTP sah dalam 10 minit

Troubleshooting covers: bot blocked by the user, wrong number, notifications disabled on the device, and iOS requiring the PWA to be installed before push works.

5.3 Matriks acara

Acara	Penerima	Saluran	Masa
Akaun diaktifkan	Staf	Emel	Segera
Tugasan baharu	Staf	Telegram + emel	Segera
Peringatan H-1	Staf	Saluran pilihan	08:00 sehari sebelum
Peringatan H-30	Staf	Saluran pilihan	30 minit sebelum mula
Resit daftar masuk	Staf	Telegram	Segera
Lewat / luar kawasan	Penyelia	Telegram + emel	Segera
Tidak hadir	Penyelia + staf	Kedua-dua	10 minit selepas tamat
Peranan penyelia diberi	Staf	Kedua-dua	Segera
Perubahan unit	Super admin	Telegram + emel	Segera
Ringkasan mingguan	Penyelia	Emel	Isnin 08:00

5.4 Templat mesej

Templates exist in both languages and are selected by the recipient's saved language.

```
BM  Telegram (tugasan baharu)
    Tugasan Luar Pejabat
    Tujuan : {tujuan}
    Lokasi : {lokasi} ({radius} m)
    Tarikh : {tarikh}, {masa_mula} - {masa_tamat}
    [ Daftar Masuk ]  [ Lihat Butiran ]

EN  Email subject: Field assignment - {date}
    You have a field assignment at {location} on {date},
    {start} to {end}. Check in from the e-HadirGo app when
    you arrive. A location check will run at check-in.
```

All templates carry the assignment id and a signed deep link, so a tap opens the correct assignment even if the message is days old.

5.5 Sandaran dan log

Delivery order is primary channel, then backup, then Gmail. Every attempt is written to notification_log with provider response, so a failure is visible rather than silent. If all channels fail, the record is flagged undelivered and the supervisor sees it on the dashboard.

6. Dwibahasa — Bilingual

Bahasa Melayu is the default; English is one tap away. Language is a property of the user, not of the device, so a shared phone in a district office still shows each person their own preference.

- **UI.** Two JSON catalogues, ms.json and en.json, with identical keys. No hardcoded strings in components.
- **Messages.** Telegram messages and emails are rendered from bilingual templates using the recipient's saved language.
- **Reports.** PDF and CSV reports render in the language of the person exporting, with the other language available as a column header option.
- **Format.** Dates as DD/MM/YYYY, times in 24-hour format, numbers with no thousands separator for years and IC-style identifiers.
- **Switching.** A switcher in the header for signed-out pages, and in Tetapan for signed-in users; the choice is cached in localStorage so the login page honours the last selection.

Malay copy is written first and English is translated from it, not the reverse, so the official record keeps the department's terminology (tugasan, kehadiran, catatan, luar kawasan).

7. PWA dan Luar Talian — PWA and Offline

Keupayaan	Tingkah laku
Pemasangan	Manifest + service worker; gesaan tambah ke skrin utama selepas lawatan kedua
Cangkang luar talian	Shell aplikasi dimuat dari cache; data terakhir dipaparkan
Daftar masuk luar talian	Disimpan dalam IndexedDB, disegerakkan apabila kembali dalam talian
Tanda sync	Rekod lewat disegerakkan ditanda synced_late dengan cap masa peranti asal
Pautan dalam	/t/<token> membuka tugasan terus dari mesej Telegram atau emel
Kamera	Swafoto pilihan semasa daftar masuk

Keupayaan	Tingkah laku
Push web	Notifikasi OS apabila dipasang; Telegram kekal sebagai saluran utama

> **Nota iOS:** Web push requires iOS 16.4 or later and the PWA installed to the home screen. Geolocation works in Safari but is stricter about permission prompts. Test on a real iPhone before UAT; do not assume parity with Android.

8. Model Data — Data Model

PostgreSQL. Eight tables carry the whole system. All timestamps are stored in UTC and rendered in Asia/Kuala_Lumpur.

users

Medan	Jenis	Nota
id	uuid	Kunci utama
email	citext, unik	Log masuk; disahkan melalui OTP
name	text	Nama penuh
password_hash	text	Argon2id
phone	text, nullable	Ditambah dari Tetapan
unit_id	uuid	Rujukan units
roles	text[]	staf, penyelia, super_admin
email_verified	boolean	Lalai false
is_dummy	boolean	Data demo atau emel tidak disahkan
language	enum	ms / en, lalai ms
is_active	boolean	Nyahaktif tanpa memadam rekod
created_at / updated_at	timestamptz	—

units

Medan	Jenis	Nota
id	uuid	Kunci utama
daerah	text	Kota Kinabalu, Sandakan, Tawau, Lahad Datu, Kudat
jabatan	text	Lalai Jabatan Perikanan Sabah
bahagian	text	Pengurusan, Perikanan Tangkapan, Akuakultur, Penguatkuasaan
unique	—	Gabungan tiga medan di atas adalah unik

assignments

Medan	Jenis	Nota
id	uuid	Kunci utama
staff_id / supervisor_id	uuid	Boleh sama jika penyelia menugaskan diri
purpose	text	Tujuan tugasan
date	date	Tarikh tugasan
start_time / end_time	time	Tetingkap masa

Medan	Jenis	Nota
location_name	text	Nama lokasi yang dilihat staf
lat / lng	numeric(9,6)	Pusat bulatan
radius_m	integer	50 – 5000
status	enum	Dijadualkan, Dalam tindakan, Selesai, Tidak hadir, Dibatalkan
created_at	timestamptz	—

attendance

Medan	Jenis	Nota
id	uuid	Kunci utama
assignment_id	uuid	Unik — satu rekod per tugasan
check_in_at / check_out_at	timestamptz	Cap masa pelayan
check_in_lat / lng	numeric(9,6)	Koordinat peranti
accuracy_m	integer	Ketepatan yang dilaporkan
distance_m	integer	Jarak ke pusat
outside_radius	boolean	Jarak > radius
outside_note	text	Wajib jika outside_radius
is_late / is_early	boolean	Mengikut jadual 4.5
late_reason	text	Wajib jika lewat
selfie_url	text, nullable	Pilihan
auto_closed	boolean	Ditutup oleh sistem
supervisor_decision	text, nullable	Terima / Tolak + nota
synced_late	boolean	Rekod luar talian yang disegerakkan

saved_locations · notification_channels · notification_log · audit_log

Jadual	Medan utama	Kegunaan
saved_locations	id, name, lat, lng, radius_m, created_by	Lokasi tetap untuk penyelia
notification_channels	user_id, type (telegram|email), identifier, verified, is_primary	Pautan saluran per pengguna
notification_log	id, user_id, event, channel, status, error, sent_at	Log penghantaran
audit_log	id, actor_id, action, entity, entity_id, before, after, created_at	Jejak audit

9. API

REST over JSON, cookie session for the web client, signed tokens for deep links and the Telegram webhook. Every endpoint enforces the role matrix in 2.2.

Kaedah	Laluan	Akses	Tujuan
POST	/api/auth/signup	Public	Cipta akaun, hantar OTP
POST	/api/auth/verify-email	Public	Sahkan OTP
POST	/api/auth/login	Public	Log masuk dengan emel + kata laluan
POST	/api/auth/logout	Auth	Tamat sesi
POST	/api/auth/forgot-password	Public	Hantar pautan set semula

Kaedah	Laluan	Akses	Tujuan
POST	/api/auth/reset-password	Public	Tetapkan kata laluan baharu
GET	/api/auth/me	Auth	Profil, peranan, unit, saluran
GET	/api/units	Auth	Senarai daerah / jabatan / bahagian
PATCH	/api/me/profile	Staf	Nama, telefon, jawatan
PATCH	/api/me/unit	Staf	Tukar unit; notifikasi admin + audit
PATCH	/api/me/password	Staf	Tukar kata laluan
PATCH	/api/me/language	Staf	Tukar bahasa
GET/PUT	/api/me/channels	Staf	Senarai dan tetapkan saluran
POST	/api/me/channels/telegram	Staf	Jana token pautan bot
POST	/api/me/channels/test	Staf	Hantar mesej ujian
GET	/api/assignments	Auth	Tugasan sendiri / unit (penyelia)
POST	/api/assignments	Penyelia	Cipta tugasan
GET	/api/assignments/{id}	Auth	Butiran + peta + rekod
PATCH	/api/assignments/{id}	Penyelia	Sunting atau batalkan
POST	/api/attendance/{id}/check-in	Staf	Daftar masuk dengan pengesahan
POST	/api/attendance/{id}/check-out	Staf	Daftar keluar
GET	/api/attendance/mine	Staf	Sejarah kehadiran sendiri
GET	/api/attendance	Penyelia	Semakan dengan penapis
PATCH	/api/attendance/{id}/review	Penyelia	Terima / tolak justifikasi
GET	/api/locations	Penyelia	Lokasi tetap
POST	/api/locations	Penyelia	Tambah lokasi tetap
GET	/api/geocode	Penyelia	Carian alamat (cache + had kadar)
GET	/api/dashboard/summary	Penyelia	Kaunter status hari ini
GET	/api/admin/users	Super admin	Senarai pengguna
PATCH	/api/admin/users/{id}/roles	Super admin	Beri / tarik penyelia
GET	/api/admin/delivery-log	Super admin	Log notifikasi
GET	/api/admin/audit	Super admin	Jejak audit
GET	/api/reports/attendance	Penyelia	CSV / JSON
GET	/api/reports/attendance.pdf	Penyelia	PDF dengan peta
POST	/api/webhooks/telegram	Secret	Terima /start dan callback butang

10. Keselamatan, Privasi & Pematuhan

Kawalan	Pelaksanaan
Pengesahan	Argon2id, sesi HttpOnly + Secure + SameSite=Lax, had kadar 5 cubaan / 15 minit
Kebenaran	Semakan peranan di pelayan pada setiap endpoint, bukan hanya di UI
Penyulitan	TLS 1.2+; pangkalan data disulitkan pada rehat
Lokasi	Diambil hanya semasa daftar masuk / keluar; tiada jejak berterusan
Persetujuan	Skrin persetujuan lokasi sebelum penggunaan pertama, boleh ditarik balik
PDPA 2010	Notis tujuan, akses, pembetulan dan penyimpanan; data lokasi dikategori peribadi
Penyimpanan	Rekod kehadiran disimpan 7 tahun; log notifikasi 90 hari; audit 7 tahun
Audit	Setiap perubahan peranan, unit dan justifikasi direkod dengan pelaku dan masa

Kawalan	Pelaksanaan
Sandaran	Harian, disimpan 30 hari, dipulihkan diuji setiap suku tahun

> **Privasi lokasi:** The system records coordinates only at check-in and check-out. It does not track movement between those points. Staff must see this stated plainly on the consent screen and in Tetapan — a system that proves presence must not become a system that watches people.

11. Had API & Kos

The core stack is free, but four services carry limits that matter at production volume.

Perkhidmatan	Kegunaan	Percuma?	Had sebenar
Telegram Bot API	Notifikasi, butang, pautan	Ya, penuh	~30 mesej/s keseluruhan, ~1 mesej/s per chat; tiada kuota bulanan
Geolocation pelayar	Koordinat daftar masuk	Ya	Tiada kuota; ketepatan 5–50 m
Leaflet + OSM	Paparan peta	Ya, fair-use	OSM awam untuk penggunaan ringan; MapTiler ~100k tile/bulan; Stadia ~200k/bulan
Nominatim	Carian alamat	Ya, polisi ketat	~1 permintaan/s, tiada autocomplete, wajib cache + atribusi
Gmail SMTP	OTP, peringatan, laporan	Ya, terhad	~500 penerima/hari (peribadi), ~2,000/hari (Workspace)
Web Push	Notifikasi PWA	Ya	Tiada had praktikal
Supabase / Neon	Pangkalan data	Ya, terhad	500 MB; projek tidur selepas ~1 minggu
Vercel / Cloudflare	Hosting	Ya, terhad	Vercel Hobby bukan komersial; cron ~2 kerja sehari
Google Maps	—	Tidak	Perlu akaun bil, bayaran per 1,000 panggilan

11.1 Apa yang pecah dahulu

At roughly 200 staff:

Risiko	Anggaran	Penyelesaian
Tile peta	~20 tile × 200 staf/hari ≈ 90k+/bulan	Peta statik semasa daftar masuk; cache tile
Emel	~600 emel/hari	Telegram untuk peringatan; Gmail untuk OTP dan rekod
Cron hosting	Peringatan H-30 perlu setiap 15 minit	cron-job.org atau VPS kecil
Pangkalan data	Projek tidur selepas tidak aktif	Pelan berbayar atau VPS

11.2 Dua konfigurasi

Konfigurasi	Komponen	Kos
A — Perintis tanpa kos	Cloudflare Pages + Workers cron, Supabase percuma, bot Telegram, Gmail Workspace, MapTiler percuma, Nominatim dengan cache	RM0 — risiko tidur, fair-use, terma bukan komersial
B — Produksi stabil	VPS 1 vCPU / 1 GB, PostgreSQL setempat, tile sendiri atau MapTiler berbayar, Amazon SES, domain .my	~RM35–40 sebulan

For a department that must keep attendance records for seven years, configuration B is the honest choice: a paused free-tier database is not an acceptable failure mode for an official record.

12. Pelan Pembinaan — Delivery Plan

Fasa	Skop	Hasil	Tempoh
P0	Asas: auth, peranan, unit, tetapan asas	Staf boleh daftar, log masuk, ubah profil	3–4 hari
P1	Tugasan + peta + lokasi tetap	Penyelia boleh cipta tugasan dengan radius	5 hari
P2	Kehadiran + peraturan masa + alert	Daftar masuk/keluar dengan pengesahan	5 hari
P3	Notifikasi Telegram + Gmail + templat	Peringatan dan alert berfungsi	4 hari
P4	Dashboard penyelia, semakan, laporan	Semakan dan eksport PDF/CSV	5 hari
P5	PWA, luar talian, dwibahasa, tetapan penuh	Boleh dipasang, BM/EN lengkap	4 hari
P6	UAT dengan data dummy, pengerasan, deploy	Sistem sedia untuk perintis	3 hari

Roughly five to six weeks for one developer. P0–P2 is the smallest useful slice: a staff member can be assigned a place and time, and check in there with evidence.

12.1 Definisi siap

- A staff member completes sign-up, verifies email, links Telegram and checks in from a phone in under three minutes.
- An out-of-radius check-in produces a warning, a required note, and a supervisor alert.
- A late check-in past the end time stores the reason and appears in the monthly report.
- Super admin grants penyelia; the user sees the new menus without re-login.
- Switching language changes UI, email template and exported report.
- Every screen works at 360 px width and on a 1280 px desktop.

13. Data Dummy — Seed Data

Phase 1 loads a demo roster so the flows can be exercised before real staff data arrives. Every seeded account carries is_dummy = true and is excluded from production reports.

13.1 Unit

Daerah	Jabatan	Bahagian
Kota Kinabalu	Jabatan Perikanan Sabah	Pengurusan
Kota Kinabalu	Jabatan Perikanan Sabah	Perikanan Tangkapan
Sandakan	Jabatan Perikanan Sabah	Akuakultur
Sandakan	Jabatan Perikanan Sabah	Penguatkuasaan
Tawau	Jabatan Perikanan Sabah	Perikanan Tangkapan
Lahad Datu	Jabatan Perikanan Sabah	Penguatkuasaan
Kudat	Jabatan Perikanan Sabah	Perikanan Tangkapan

13.2 Pengguna

Nama	Emel	Unit	Peranan

Nama	Emel	Unit	Peranan
Administrator Sistem	admin.hadirgo@sabah.gov.my	KK / Pengurusan	super_admin
Mohd Faizal bin Osman	faizal.osman@sabah.gov.my	KK / Perikanan Tangkapan	staf + penyelia
Nurul Aisyah binti Rahman	nurul.aisyah@sabah.gov.my	KK / Perikanan Tangkapan	staf
Jacqueline anak Jantan	jackie.jantan@sabah.gov.my	Sandakan / Akuakultur	staf
Ahmad Zulkifli bin Hamzah	zulkifli.hamzah@sabah.gov.my	Sandakan / Pengurusan	staf + penyelia
Siti Khadijah binti Yusof	khadijah.yusof@sabah.gov.my	Tawau / Perikanan Tangkapan	staf
Lim Wei Sheng	weisheng.lim@sabah.gov.my	Tawau / Akuakultur	staf
Dayang Norhayati binti Awang	norhayati.awang@sabah.gov.my	Lahad Datu / Penguatkuasaan	staf + penyelia
Ronaldo anak Belayan	ronaldo.belayan@sabah.gov.my	Kudat / Perikanan Tangkapan	staf
Pengguna Dummy	dummy@dummy.test	KK / Pengurusan	staf (is_dummy)

Kata laluan demo untuk semua akaun: HadirGo#2026

13.3 Lokasi tetap

Nama	Latitud	Longitud	Radius
Jeti Kota Kinabalu	5.980400	116.073500	250 m
Pejabat Perikanan Daerah Sandakan	5.840200	118.117900	150 m
Pusat Inkubasi Tuaran	6.121000	116.221000	200 m
Stesen Perikanan Tawau	4.244800	117.891200	250 m

14. Log Keputusan & Item Terbuka

14.1 Keputusan

ID	Keputusan
D-01	Sistem tertutup untuk staf; akaun aktif automatik dengan @sabah.gov.my
D-02	Log masuk emel + kata laluan; IC dikeluarkan dari aliran pengesahan
D-03	Semua akaun baharu berperanan staf; penyelia diberi oleh super admin
D-04	Penyelia boleh menugaskan diri sendiri untuk kerja luar
D-05	Lokasi ditetapkan sebagai pin + bulatan radius 50 m – 5 km
D-06	Daftar masuk awal dibenarkan pada bila-bila masa sebelum masa mula
D-07	Lewat selepas 15 minit; sebab wajib untuk setiap rekod lewat
D-08	Daftar masuk selepas masa tamat dibenarkan dengan sebab wajib kepada penyelia
D-09	Luar radius: amaran sebelum hantar + catatan wajib
D-10	Swafoto pilihan, bukan wajib
D-11	Saluran: Telegram dan Gmail sahaja; tiada WhatsApp

ID	Keputusan
D-12	Dwibahasa BM/EN, boleh ditukar dalam Tetapan
D-13	Perubahan unit serta-merta, dengan notifikasi admin dan audit
D-14	Data dummy untuk fasa pertama
D-15	Pengesahan emel melalui MX + OTP; laluan dummy dengan pengesahan
D-16	Web responsif + PWA, satu pangkalan kod
D-17	Hanya daftar masuk/keluar; tiada jejak berterusan
D-18	Peta statik semasa daftar masuk; peta interaktif pada dashboard

14.2 Item terbuka

#	Item	Cadangan
1	Masa sebenar peringatan H-1 dan H-30	H-1 08:00; H-30 tepat 30 minit sebelum mula
2	Tempoh penyimpanan rekod	7 tahun untuk rekod kehadiran, 90 hari untuk log notifikasi
3	Radius lalai	250 m, boleh diubah per tugasan
4	Penyelia menyunting rekod selepas semakan	Tidak; hanya justifikasi, dengan audit
5	Sokongan berbilang bahasa selain BM/EN	Tidak untuk v1; struktur i18n sedia untuk tambahan

15. Lampiran — Screen Inventory

Kawasan	Skrin	Peranan
Awam	Halaman utama, Daftar, Log masuk, Lupa kata laluan	Semua
Onboarding	Pengesahan emel, Pilih saluran, Panduan Telegram	Staf
Staf	Dashboard, Butiran tugasan, Daftar masuk, Daftar keluar	Staf
Staf	Rekod saya, Tetapan, Panduan notifikasi	Staf
Penyelia	Tugasan baharu, Senarai tugasan, Peta langsung	Penyelia
Penyelia	Semakan, Butiran rekod, Laporan	Penyelia
Super admin	Pengguna & Peranan, Unit, Lokasi tetap	Super admin
Super admin	Templat mesej, Log penghantaran, Audit	Super admin

— Tamat dokumen —